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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

EC2001-1 – ANALOG ELECTRONIC CIRCUITS-I
(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

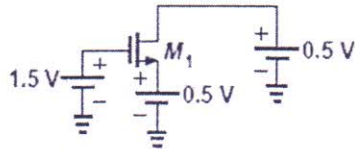
Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

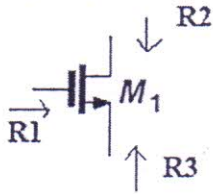
20 Marks

- Which of the following describes the most desirable operating point for small-signal amplification?
A) A point near cutoff to maximize negative signal swing.
B) A point near saturation to maximize positive signal swing.
C) A point where the spacing between IB curves is rapidly changing, indicating high gain.
D) A region of more linear spacing, allowing for linear gain and the largest possible voltage and current swing without entering cutoff or saturation.
- The voltage from the emitter to ground (V_E) in an emitter-bias configuration is determined by which formula?
A) $V_E = V_{CC} - I_C R_C$
B) $V_E = V_{BE}$
C) $V_E = I_E R_E$
D) $V_E = V_{CE} + V_{CC}$
- For a voltage divider bias circuit, what are the coordinates of end points of DC load line?
A) $(V_{CC}, 0)$ $(0, \frac{V_{CC}}{R_C})$
B) $(V_{CC}, 0)$ $(0, \frac{V_{CC}}{R_C + R_E})$
C) $(V_{CC}, 0)$ $(0, \frac{V_{CC}}{R_B})$
D) $(0, V_{CC})$ $(\frac{V_{CC}}{R_B}, 0)$
- Which of the following is not correct statement in the approximate analysis of a voltage divider bias circuit?
A) $\beta r_e \geq 10R_2$
B) R_1 and R_2 are considered to be as series elements
C) I_B is essentially zero
D) $\beta R_E \geq 10R_2$
- For a transistor biased at $I_E = 2 \text{ mA}$, the value of r_e (at 300 K) is approximately:
A) 13Ω
B) 52Ω
C) 26Ω
D) 100Ω
- The Early effect occurs due to:
A) Forward biasing of the base-emitter junction
B) Widening of the base due to high current
C) Narrowing of the base width with increase in V_{CE}
D) Reduction of collector doping
- If the emitter bypass capacitor opens, the AC output voltage will
A) Remain the same
B) Decrease
C) Increase
D) Equal to zero
- If the base voltage in an emitter follower is 5 V, the emitter voltage will be approximately:
A) 5
B) 0.7
C) 0
D) 4.3

9. The P-MOSFET is made up of
 A) N-type source, N-type drain and P-type bulk B) N-type source, P-type drain and P-type bulk
 C) P-type source, N-type drain and N-type bulk D) P-type source, P-type drain and N-type bulk
10. Triode region of N-MOSFET is defined by following voltage conditions
 A) $V_{GS} > V_{TH}$, and $V_{DS} \geq V_{GS} - V_{TH}$ B) $V_{GS} < V_{TH}$
 C) $V_{GS} > V_{TH}$, and $V_{DS} < V_{GS} - V_{TH}$ D) $V_{GS} < V_{TH}$, and $V_{DS} < V_{GS} - V_{TH}$
11. Determine the region of operation of M1 in the circuit shown in Fig. Assume $V_{TH}=0.4V$



- A) Cut-off region B) Triode region
 C) Saturation region D) Sub threshold conduction
12. Choose the correct statement
 A) MOSFET is a unipolar, voltage controlled, two terminal device B) MOSFET is a bipolar, current controlled, three terminal device
 C) MOSFET is a unipolar, voltage controlled, three terminal device D) MOSFET is a bipolar, current controlled, two terminal device
13. The channel length of an NMOSFET is defined as the distance between:
 A) Source and drain B) Gate and substrate
 C) Gate and source D) Gate and drain
14. R_1 , R_2 , R_3 for the NMOSFET shown in the figure are



- A) $R_1 = \frac{1}{g_m}, R_2 = r_o, R_3 = \infty$ B) $R_1 = \infty, R_2 = r_o, R_3 = \frac{1}{g_m}$
 C) $R_1 = r_o, R_2 = \infty, R_3 = \frac{1}{g_m}$ D) $R_1 = \infty, R_2 = \frac{1}{g_m}, R_3 = r_o$
15. For the MOSFET operating in saturation, if the channel length is doubled, drain current I_D changes by a factor
 A) 0.5 B) 4
 C) 2 D) No change
16. If a MOSFET is to be used in the making of an amplifier then it must work in
 A) Cut-off region B) Triode region
 C) Saturation region D) Both cut-off and triode region can be used
17. The main requirement for accurate current mirroring is:
 A) Large channel length difference between MOSFETs B) High drain resistance
 C) Different substrate connections D) Both MOSFETs must have identical threshold voltages and aspect ratios
18. The current source connected to the common source node is used to:
 A) Provide load resistance B) Bias the transistors and set tail current
 C) Stabilize the gate voltage D) Reduce drain current
19. The differential amplifier rejects:
 A) Differential signals B) AC signals
 C) Common-mode signals D) DC signals
20. The type of power amplifier which exhibits crossover distortion in its output is
 A) Class A B) Class B
 C) Class C D) Class AB

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Determine the range of possible values for V_C for the network of Figure. Q1(a) using the $1M\Omega$ potentiometer. Assume $V_{BE}=0.7V$.	8	L*3	1	1
b)	For Emitter follower configuration, draw the small signal circuit. Derive the expression for input impedance, output impedance and voltage gain neglecting Early Effect.	8	L2	2	1
2. a)	Design a voltage-divider bias network using a supply of 24 V, a transistor with a beta of 110, and an operating point of $I_{CQ} = 4$ mA and $V_{CEQ} = 8$ V. Choose $V_E = 1/8V_{CC}$.	8	L3	1	1
b)	For emitter unbypassed CE amplifier voltage divider bias circuit, draw the small signal circuit, derive the expression for the input impedance, output impedance and voltage gain using re model of transistor. Neglect the effect of r_o .	8	L2	2	1
3. a)	Determine the Q-point values of I_C and V_{CE} for the circuit in Figure Q3(a) Find $I_{C(sat)}$ and $V_{CE(cut\ off)}$, and then construct the DC load line and plot the Q-point. Assume $I_C \approx I_E$ to find $I_{C(sat)}$ and $V_{CE(cut\ off)}$	8	L3	1	1
b)	For the network of Figure.Q3b : i) Calculate I_B and I_C ii) Determine r_e iii) Determine Z_i and Z_o iv) Find A_v .	8	L3	2	1
Unit – II					
4. a)	With the help of volt-ampere characteristic, explain the operating regions of enhancement-type NMOS transistor. Also, give the expression for drain current in each operating region.	8	L2	3	1
b)	Draw a neat circuit diagram of a Common Source stage with degeneration and voltage divider biasing. Draw its small signal model and obtain the expression for small signal gain, input impedance and output impedance considering $\lambda=0$.	8	L2	4	1
5. a)	In the circuit of Figure Q5a , determine maximum transconductance that M1 can provide without going into the triode region. Assume $V_{TH} = 0.4V$ and $\lambda = 0$.	8	L3	3	1
b)	For the circuit shown in Figure Q5b , draw the small signal model and obtain the expression for voltage gain A_v , input impedance R_{in} considering $\lambda=0$.	8	L2	4	1
6. a)	Draw a neat circuit diagram of a Source follower core. Draw its small signal model and obtain the expression for small signal gain, input and output impedance considering $\lambda \neq 0$.	8	L2	4	1
b)	In the circuit of Figure.Q6b , determine the maximum allowable value of W/L if M1 must remain in saturation. Assume $V_{TH} = 0.4V$, $\mu_n C_{ox} = 200 \mu A/V^2$, and $\lambda = 0$.	8	L3	3	1
Unit – III					
7. a)	Draw the circuit diagram of MOS differential pair. Explain with relevant equations and waveforms common mode and differential mode operation.	8	L2	5	1
b)	Explain with relevant circuit diagram and equations MOS Current-Steering Circuits. Calculate I_{copy} in the circuit shown in Figure Q7(b) Assume all of the transistors operate in saturation.	8	L3	5	1

8. a) Draw the circuit diagram of MOS differential pair. Derive the expression for small signal differential gain.
 b) Explain with relevant circuit diagram and waveforms operation of class A output stage.

8	L2	5	1
8	L2	5	1

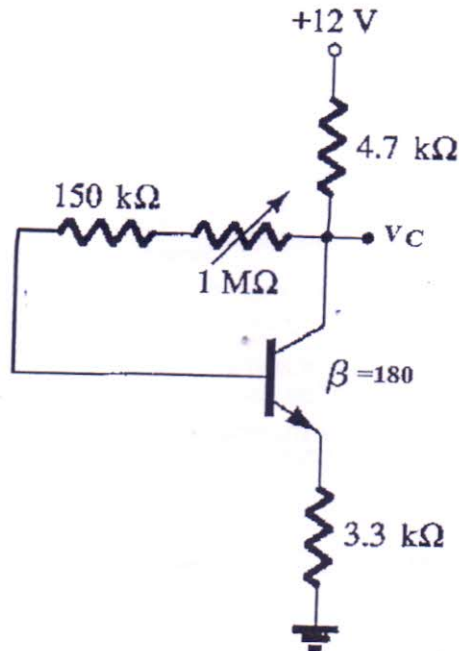


Figure Q1a

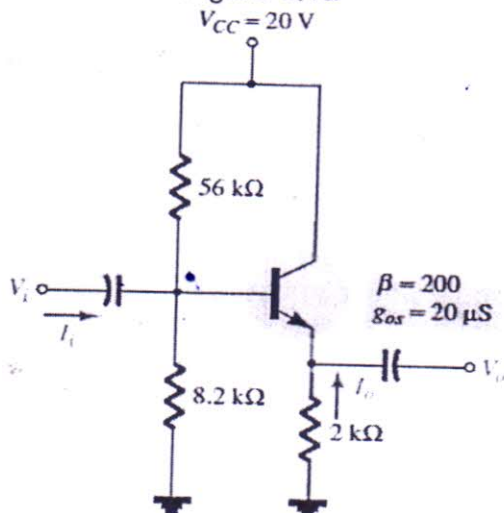


Figure Q3b

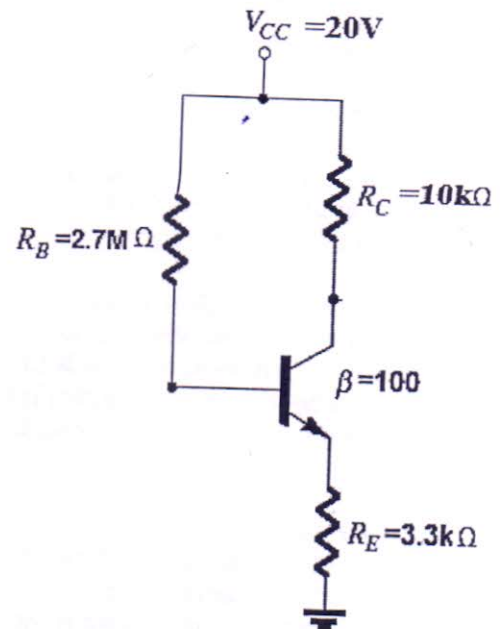


Figure Q3a

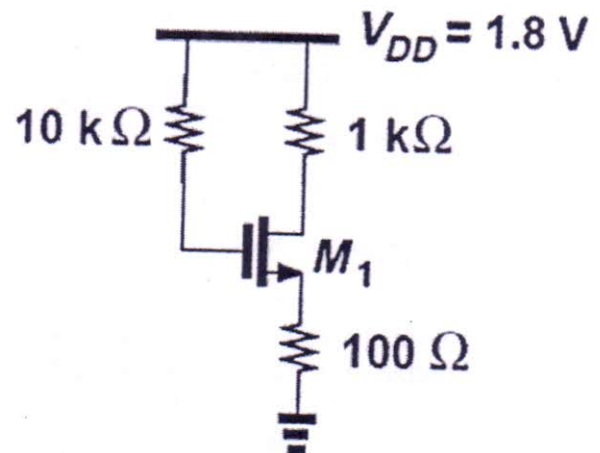


Figure Q5a

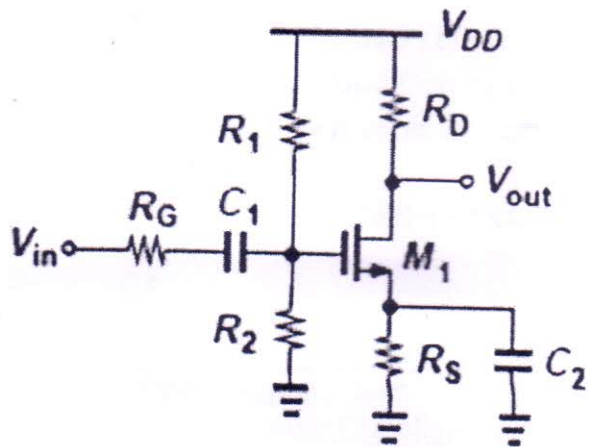


Figure. Q5b

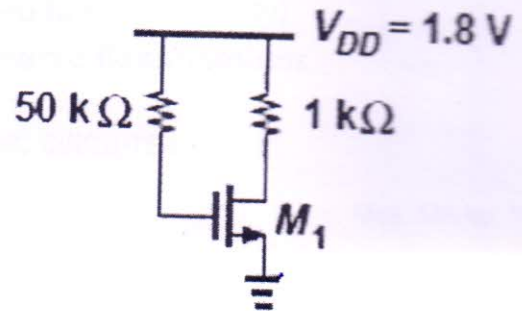


Figure. Q6b

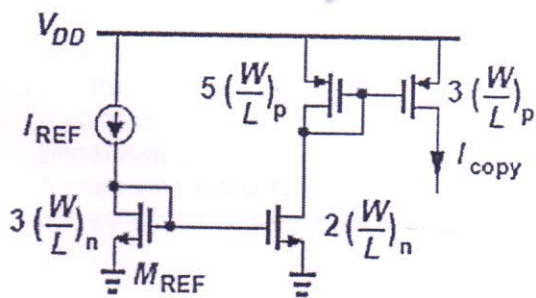


Figure. Q7b

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
